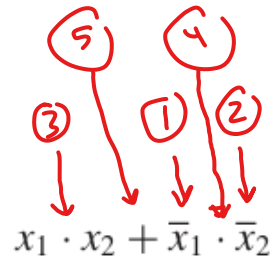


Precedence of operations:

1. NOT
2. AND
3. OR

- Thus, in the expression



First, we must generate the complements of x_1 and x_2 .

Then the product terms $x_1 \cdot x_2$ and $x_1' \cdot x_2'$ are formed.

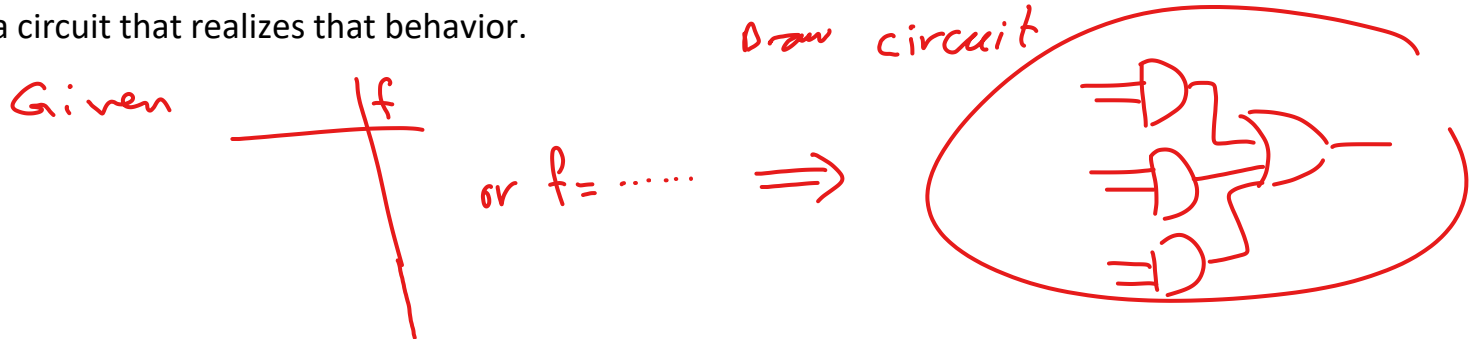
Followed by (+) operation.

You can use parentheses to ensure that order as follows:

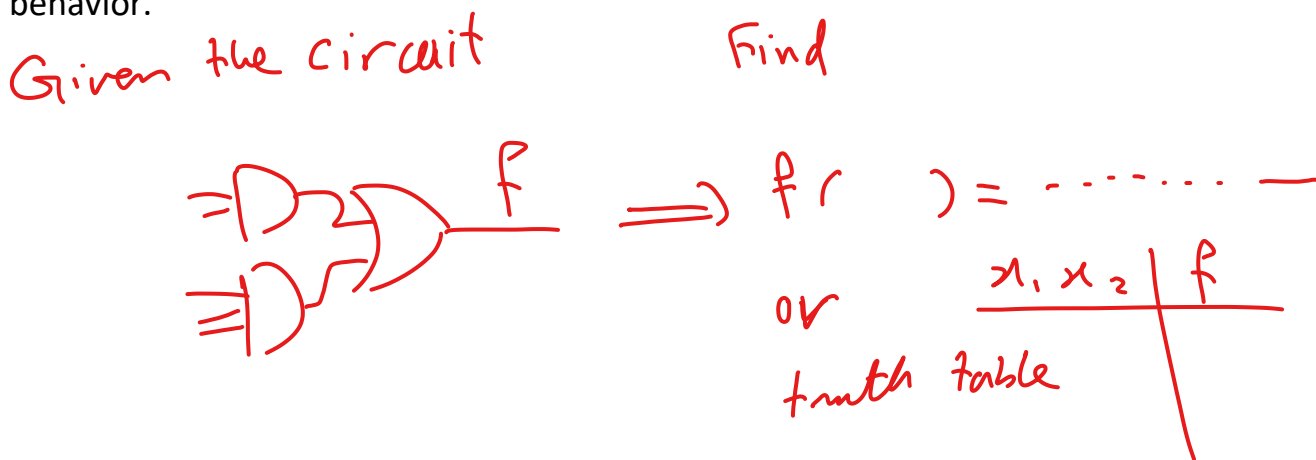
$$(x_1 \cdot x_2) + ((\bar{x}_1) \cdot (\bar{x}_2))$$

- Throughout this course, we will work with two fundamental types of tasks. Every question we study will belong to one of these two types:

1- Synthesis process: Given a description of the desired functional behavior, you then generate a circuit that realizes that behavior.



2- Analysis process: Given a circuit (a network of logic gates), you determine its functional behavior.

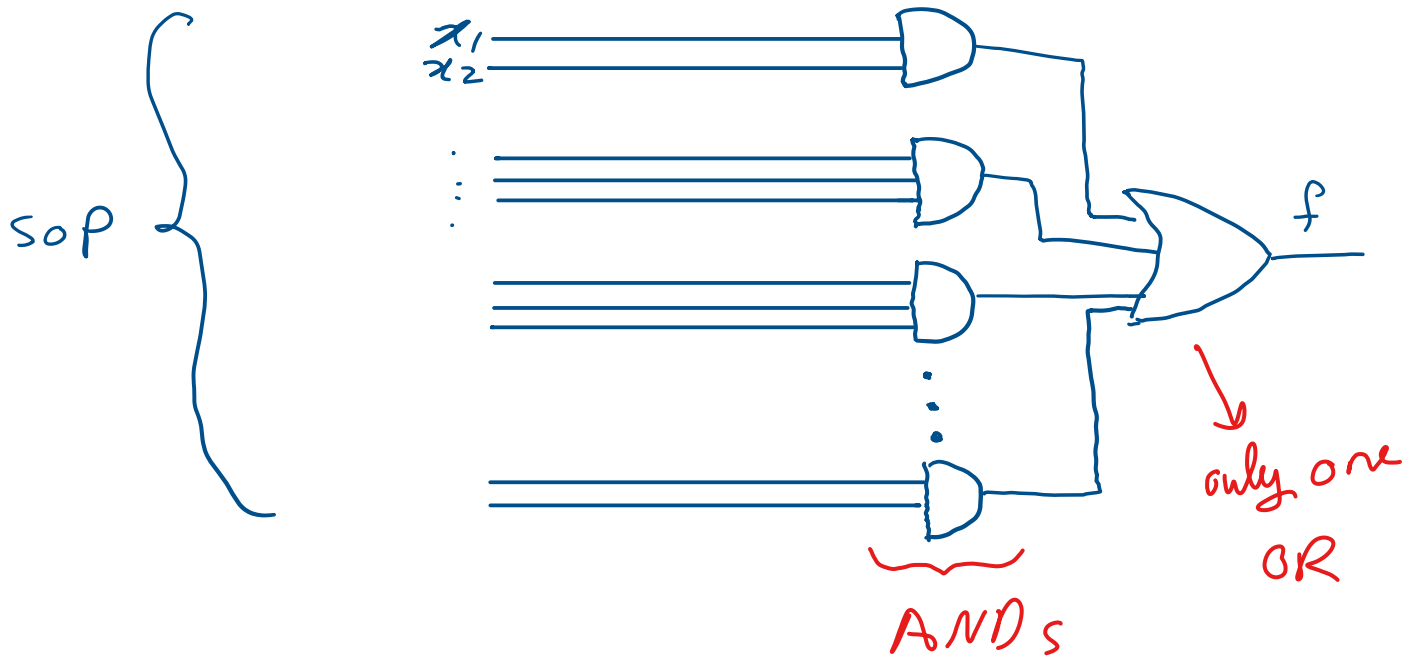


- Any two-level network of gates (circuit) can be represented in two forms:

1- Sum of Products (SOP)

2- Products of Sums (POS)

- Sum of Products (SOP): In general, it looks like the following circuit.



$$f(x_1, x_2, x_3) = \underbrace{x_1 \bar{x}_2 \bar{x}_3}_{\text{minterms}} + \underbrace{\bar{x}_1 x_2 \bar{x}_3}_{\text{ANDs}} + \underbrace{\bar{x}_1 x_2}_{\text{ANDs}} + \underbrace{x_2 x_3}_{\text{ANDs}} + \dots$$

sop

minterms

ANDs

- Draw a truth table for the following SOP expression:

$$f(x_1, x_2, x_3) = \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2$$

* Truth table

	x_1	x_2	x_3	f
$m_0 \rightarrow 0$	0	0	0	1
$m_1 \rightarrow 0$	0	0	1	1
$m_2 \rightarrow 0$	0	1	0	1
m_3	0	1	1	0
m_4	1	0	0	0
m_5	1	0	1	1
m_6	1	1	0	1
m_7	1	1	1	0

$$\text{SOP} \rightarrow f = 1$$

$$x \leftrightarrow 1$$

$$\bar{x} \leftrightarrow 0$$

SOP

$$f(x_1, x_2, x_3) = \bar{x}_1 \bar{x}_2 \bar{x}_3 + \bar{x}_1 \bar{x}_2 x_3 + \bar{x}_1 x_2 \bar{x}_3 + x_1 \bar{x}_2 x_3 + x_1 x_2 \bar{x}_3$$

$$= m_0 + m_1 + m_2 + m_5 + m_6$$

$$= \sum m(0, 1, 2, 5, 6)$$

Canonical
SOP

- A **minterm** is an **AND** (or product) term that includes **every** variable exactly once, either in its true form (x) or complemented form (x').

- **Canonical SOP:** Is an SOP in which **every term** is a **minterm**.

- In other words, it is a sum of minterms.

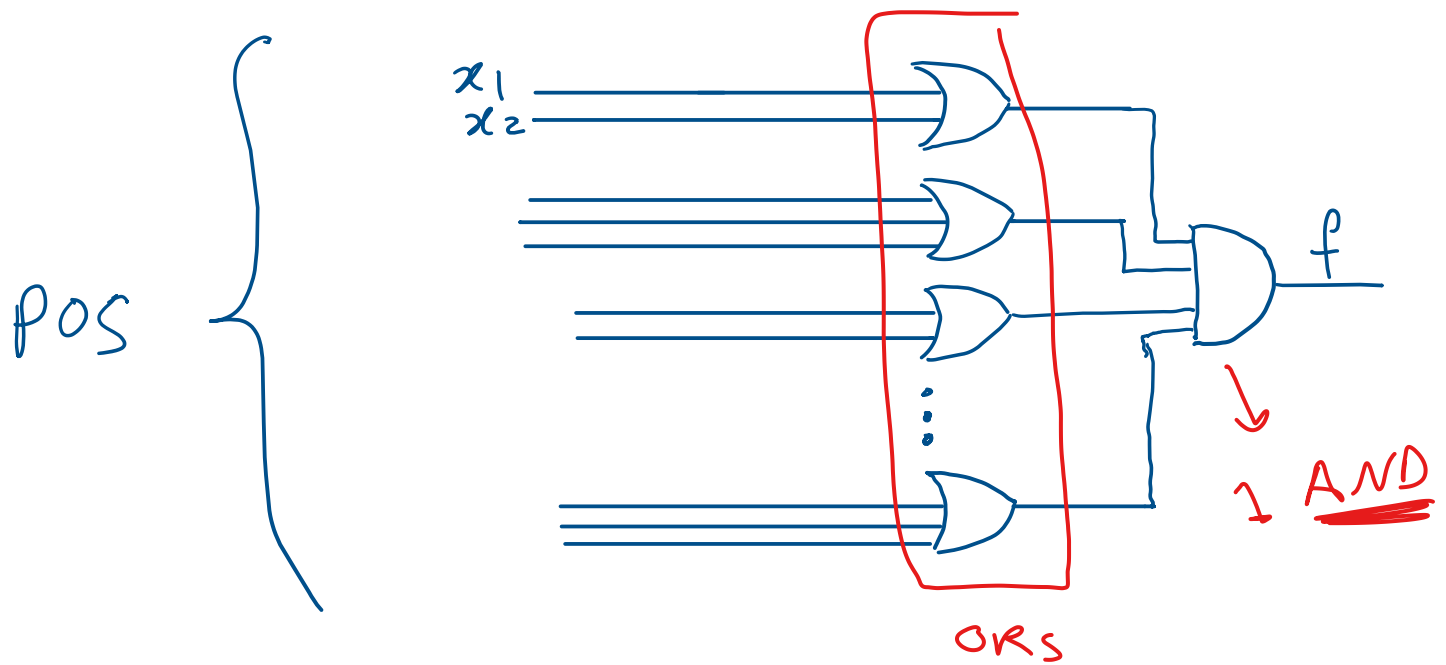
$$\begin{array}{ccc} 4 & 2 & 1 \\ x_1 & \bar{x}_2 & x_3 = 101 = m_5 \end{array}$$

$$\begin{array}{ccc} 4 & 2 & 1 \\ a & b & \bar{c} = 110 = m_6 \end{array}$$

$$\begin{array}{ccc} 4 & 2 & 1 \\ m_3 = & \bar{x}_1 & x_2 x_3 \end{array}$$

$$\begin{array}{cccc} 8 & 4 & 2 & 1 \\ m_{12} = & x_1 & x_2 & \bar{x}_3 \bar{x}_4 \\ & 1 & 1 & 00 \end{array}$$

- Product of Sums (POS): In general, it looks like the following circuit.



$$f(x_1, x_2, x_3) = (x_1 + \bar{x}_2 + x_3) \cdot (\bar{x}_1 + \bar{x}_2 + x_3) \cdot (x_1 + x_2 + x_3) \cdot (\bar{x}_2 + x_3) \cdot \dots$$

pos

Maxterms

ORs

The equation shows the function $f(x_1, x_2, x_3)$ as a product of sum terms. Red arrows point from each sum term to the label "Maxterms". Another set of red arrows points from each sum term to the label "ORs", indicating that each sum term is implemented by an OR gate. The label "pos" is written in blue with an arrow pointing to the equation.

- Draw a truth table for the following POS expression:

$$g(x_1, x_2, x_3) = (\bar{x}_1 + x_2 + x_3) \cdot (\bar{x}_2 + \bar{x}_3)$$

x_1	x_2	x_3	g
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

$$POS \Rightarrow g = 0$$

$$\bar{x} \longleftrightarrow 1$$

$$x \longleftrightarrow 0$$

POS

$$g(x_1, x_2, x_3) = (x_1 + \bar{x}_2 + \bar{x}_3) \cdot (\bar{x}_1 + x_2 + x_3) \cdot (\bar{x}_1 + \bar{x}_2 + \bar{x}_3)$$

$$= M_3 \cdot M_4 \cdot M_7$$

$$= \prod M(3, 4, 7)$$

Canonical
POS

A **Maxterm** is an **OR** (or sum) term that includes **every** variable exactly once, either in its true form (x) or complemented form (x').

Canonical POS: Is an POS in which **every term** is a **Maxterm**.

- In other words, it is a product of Maxterms.

$$\begin{array}{cccc} & 4 & 2 & 1 \\ M_3 = & 0 & 1 & 1 \end{array} = x_1 + \bar{x}_2 + \bar{x}_3$$

$$\begin{array}{cccc} & 4 & 2 & 1 \\ M_5 = & \bar{x}_1 & x_2 & \bar{x}_3 \end{array}$$

$$\begin{array}{cccc} 8 & 4 & 2 & 1 \\ \bar{a} + b + \bar{c} + \bar{d} = M_{11} \end{array}$$

Note that from previous truth tables, $f = g$

$$f = g \Rightarrow \sum m(0, 1, 2, 5, 6) = \prod M(3, 4, 7)$$


_Derive the canonical SOP for f' (i.e. $\bar{f}$)

x_1	x_2	x_3	f	$\bar{f}$
0	0	0	1	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	0	1

Diagram showing the derivation of the canonical SOP for $\bar{f}$. The truth table shows that $\bar{f}$ is 1 for minterms 3, 4, and 7. These minterms are circled in red, and arrows point from them to a circle labeled "SOP".

$$\begin{aligned}
 \bar{f}(x_1, x_2, x_3) &= \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 x_2 x_3 \\
 &= m_3 + m_4 + m_7 \\
 &= \sum m(3, 4, 7)
 \end{aligned}$$

$$f = \sum m(0, 1, 2, 5, 6) = \prod M(3, 4, 7)$$

$$\bar{f} = \sum m(3, 4, 7)$$

Example: Given the logical expression $f(a, b, c) = \sum m(1, 3, 5, 6)$,

- 1- Find the canonical POS form for f .
- 2- Find the canonical SOP form for f' (i.e. the complement of f)
- 3- Determine the simplest SOP form for f .

$$1) f(a, b, c) = \prod M(0, 2, 4, 7)$$

$$2) \bar{f}(a, b, c) = \sum m(0, 2, 4, 7)$$

$$3) f(a, b, c) = \sum m(1, 3, 5, 6)$$
$$= m_1 + m_3 + m_5 + m_6$$

$$= \bar{a}\bar{b}c + \bar{a}bc + a\bar{b}c + ab\bar{c}$$

$$= \bar{a}c + a\bar{b}c + ab\bar{c}$$

$$= c(\bar{a} + a\bar{b}) + ab\bar{c}$$

$$= c(\bar{a} + \bar{b}) + ab\bar{c}$$

$$= c\bar{a} + c\bar{b} + ab\bar{c}$$